# Problem E. Flower's land 3

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|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 1 second |
| Memory limit: | 1024 megabytes |

There is a binary string $s_1$ of length $m$ stored on disk #1.

Then, there are $n-1$ operations. In the $i$-th operation, a disk $1 \le p_{i+1} \le i$ is chosen, and the string $s_{p_{i+1}}$ on that disk is copied to disk $i+1$, producing $s_{i+1}$. However, during the copying process, up to $k$ bits may be flipped (i.e., errors occur in at most $k$ positions).

You are given all the final $n$ binary strings $s_1, s_2, \ldots, s_n$, each of length $m$. Your task is to determine how many possible sequences $p_2, p_3, \ldots, p_n$ could result in this final configuration.

Since the answer can be large, you only need to find the answer modulo 998244353.

## Input

The first line of the input contains three integers $n, m, k$ ($2 \le n \le 5000, 4 \le m \le 15000, 1 \le k \le 3$), described in the statement. It is guaranteed that $m$ is a multiple of 4.

Each of the next $n$ lines contains a hexadecimal string $s'_i$ of length $m/4$. Each character of $s'_i$ is one of `0-9` or `A-F`, where `A` $= 10$, `B` $= 11$, $\ldots$, `F` $= 15$.

Each bit in the hexadecimal representation $s'_i$ corresponds to four consecutive bits in the binary string $s_i$. Specifically, for each bit $s'_{i,j}$, it can be proved that there exists a unique tuple $(a, b, c, d)$ satisfying $s'_{i,j} = 8a + 4b + 2c + d$ and $a, b, c, d \in \{0, 1\}$. Bits in the binary string $s_i$ satisfy $(s_{i,4j}, s_{i,4j+1}, s_{i,4j+2}, s_{i,4j+3}) = (a, b, c, d)$.

## Output

Print an integer — the number of valid sequences $(p_2, p_3, \ldots, p_n)$ modulo 998244353.

## Example

| standard input | standard output |
|---|---|
| 5 8 2<br>95<br>05<br>BD<br>9C<br>BD | 6 |